



机器人 + 平台
Robot + Platform



适配型号 Compatible Models

可搭配磁吸式底座，适配（FR3、FR3WML、FR5、FR5WML、FR10）。
It can be equipped with a magnetic suction base, compatible with models such as（FR3、FR3WML、FR5、FR5WML、FR10）.

车载式气保焊套装应用案例 Application Case of Vehicle-mounted MIG Welding Kit



协作机器人搭载移动小车，集成焊接设备与快速编程手柄，打造灵活高效的移动式焊接解决方案。

核心优势：无需固定安装，可快速部署至不同工位；协作机器人保障人机协同安全，快速编程手柄简化示教流程。

典型场景：适用于工程机械、钢结构等行业的大尺寸工件现场焊接、补焊及复杂焊缝作业，大幅提升焊接柔性 with 生产效率。

The collaborative robot is mounted on a mobile cart, integrating welding equipment and a quick-programming handle to deliver a flexible and efficient mobile welding solution.

Core Advantages: No fixed installation is required, enabling rapid deployment to different workstations; the collaborative robot ensures human-machine collaborative safety, and the quick-programming handle simplifies the teaching process.

Typical Scenarios: Suitable for on-site welding, repair welding and complex seam operations of large workpieces in industries such as engineering machinery and steel structures, greatly improving welding flexibility and production efficiency.

标准车体气保焊套装应用案例 Application Case of Standard Mobile Cart MIG/MAG Welding System



协作机器人搭载标准移动车体，集成气保焊枪、送丝机构与保护气体钢瓶，打造灵活高效的自动化焊接解决方案。

核心优势：无需固定安装，可快速部署至不同工位；协作机器人保障人机协同安全，柔性适配多种规格工件，实现高效自动化焊接作业。

典型场景：适用于 CNC 加工中心、数控车床、磨床等机床的自动上下料作业，可替代人工完成工件取放、装夹与转运，大幅提升生产效率与设备稼动率，降低人工成本。

The collaborative robot is mounted on a standard mobile cart, integrated with MIG/MAG welding torch, wire feeder and protective gas cylinder, to create a flexible and efficient automatic welding solution.

Core Advantages: No fixed installation is required, enabling rapid deployment to different workstations; the collaborative robot ensures human-machine collaborative safety, flexibly adapts to workpieces of various specifications, and realizes efficient automatic MIG/MAG welding operations.

Typical Scenarios: Suitable for on-site welding, repair welding and complex seam operations of large workpieces in industries such as engineering machinery and steel structures, greatly improving welding flexibility and production efficiency.



适配型号 Compatible Models

可搭配磁吸式底座，适配（FR3WML、FR5）。
It can be equipped with a magnetic suction base, compatible with (FR3WML, FR5).



适配型号 Compatible Models

可搭配磁吸式底座，适配（FR3WML、FR5、FR5WML、FR10）。
It can be equipped with a magnetic suction base, compatible with (FR3WML, FR5, FR5WML, FR10).

轻型轨道焊接套装应用案例 Application Case of Light Rail Welding System



协作机器人搭载轻型可拼接轨道，打造灵活高效的自动化焊接解决方案。轨道支持 3-4 米延长拓展，能适应更长工件与更大作业范围，满足多样化焊接场景需求。

核心优势：无需固定安装，可快速切换工位部署；伺服电机与齿轮齿条驱动保障精准平稳运动，开关磁铁实现快速锁止固定；协作机器人保障人机协同作业安全，柔性适配多规格工件，实现稳定高效的自动化焊接。

典型场景：适用于金属结构件、钣金工件的自动化弧焊作业，可替代人工完成长距离焊缝与复杂位置焊接，有效提升生产效率与设备稼动率，降低人工成本。

The collaborative robot is mounted on a light-duty spllicable rail to create a flexible and efficient automatic welding solution. The rail supports 3-4 meters of extension to adapt to longer workpieces and larger operating ranges, meeting diverse welding scenario requirements.

Core Advantages: No fixed installation is required, enabling quick deployment across different workstations; servo motors and rack-and-pinion drive ensure precise and smooth movement, while switch magnets enable quick locking and fixation; the collaborative robot ensures safe human-machine collaboration, flexibly adapts to workpieces of various specifications, and delivers stable and efficient automatic welding.

Typical Scenarios: Suitable for automatic arc welding of metal structural parts and sheet metal workpieces. It can replace manual work to complete long-distance welds and complex-position welding, effectively improving production efficiency and equipment utilization, while reducing labor costs.

龙门架倒挂 3D 视觉焊接应用案例 Gantry-Mounted Inverted 3D Vision Welding Application Case



轻型龙门架倒挂协作机器人，集成视觉定位相机与水冷焊枪，打造高精度、柔性化的自动焊接解决方案。

核心优势：无需固定安装，可快速部署至不同工位；视觉相机实现工件精准定位与焊缝寻位，协作机器人保障人机协同安全，实现高效自动化焊接作业。

典型场景：适用于工程机械、钢结构、钣金件等行业的工件焊接作业，可替代人工完成高精度焊缝焊接、补焊及复杂焊缝作业，大幅提升焊接柔性、生产效率与焊缝质量。

The light gantry inverts the collaborative robot, integrating a visual positioning camera and a water-cooled welding torch, to create a high-precision, flexible automatic welding solution.

Core Advantages: No fixed installation is required, enabling rapid deployment to different workstations; the vision camera realizes accurate workpiece positioning and seam tracking, the collaborative robot ensures human-machine collaborative safety, and achieves efficient automatic welding operations.

Typical Scenarios: Suitable for workpiece welding operations in industries such as engineering machinery, steel structures and sheet metal parts. It can replace manual work to complete high-precision seam welding, repair welding and complex seam operations, greatly improving welding flexibility, production efficiency and weld quality.



适配型号 Compatible Models

适配 (FR3WML、FR5、FR5WML、FR10)。
Compatible with (FR3WML, FR5, FR5WML, FR10).



适配型号 Compatible Models

激光焊接套装可适配FR10、FR20系列协作机器人。小车本体可适配多种型号协作机器人，可搭配磁吸底座。
The laser welding kit is compatible with FR10 and FR20 series collaborative robots. The cart body is compatible with various models of collaborative robots and can be equipped with a magnetic suction base.

车载式激光焊接套装应用案例 Application Case of Vehicle-mounted Laser Welding Kit



协作机器人集成激光焊接系统与移动小车平台，配合快速编程手柄，实现高精度、高柔性的移动式激光焊接作业。

核心优势: 无需固定工位，可快速奔赴生产现场开展作业；激光焊接热影响区小，焊缝美观牢固；协作模式保障人机共线安全，快速编程大幅缩短调试时间。

典型场景: 适用于汽车车身、精密钣金、工程机械结构件等领域的现场拼接焊、精密补焊及异形焊缝加工，有效解决大件搬运难题，提升生产效率与焊接质量。

The collaborative robot integrates a laser welding system and a mobile cart platform, paired with a quick-programming handle, to achieve high-precision and flexible mobile laser welding operations.

Core Advantages: No fixed workstation is needed, enabling rapid deployment to production sites; laser welding features a small heat-affected zone with aesthetic and firm welds; collaborative mode ensures human-machine co-line safety, and quick programming greatly shortens debugging time.

Typical Scenarios: Suitable for on-site butt welding, precision repair welding and special-shaped seam processing in fields such as automotive body, precision sheet metal and engineering machinery structural parts, effectively solving the problem of large workpiece handling and improving production efficiency and welding quality.



适配型号 Compatible Models

适配 (FR3WML、FR5、FR5WML、FR10)。
Compatible with (FR3WML, FR5, FR5WML, FR10).

汽车零部件上下料应用案例

Application Case of Automotive Parts Loading & Unloading



协作机械臂集成高负载轻量化关节模组、柔性夹爪、智能视觉系统等核心模块，可在汽车零部件制造场景中完成高精度上下料、转运与装配作业，实现产线自动化升级。

核心优势：搭载高负载轻量化关节模组，兼顾性能与便捷；机械臂灵活高效，运动覆盖广；集成视觉感知与智能控制系统，可精准识别工件、自主规划路径，模块化设计支持快速定制，适配汽车零部件产线节拍。

典型场景：适用于汽车零部件智能制造车间，可完成发动机部件、车身钣金件、电子元器件等上下料、转运与精密装配任务，实现 24 小时自动化作业，提升产线效率，降低人工成本。

The collaborative robotic arm integrates core modules such as high-load lightweight joint modules, flexible grippers, and intelligent vision systems, enabling high-precision loading, unloading, transfer and assembly operations in automotive parts manufacturing scenarios, realizing production line automation upgrading.

Core Advantages: Equipped with high-load lightweight joint modules, balancing performance and convenience; the robotic arm is flexible and efficient with wide motion coverage; integrates visual perception and intelligent control systems, enabling accurate workpiece recognition and autonomous path planning, with modular design supporting rapid customization to adapt to automotive parts production line rhythm.

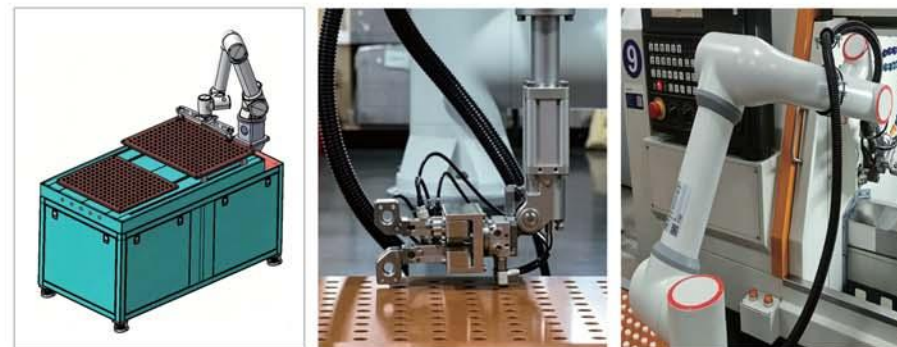
Typical Scenarios: Suitable for smart manufacturing workshops of automotive parts, completing tasks such as loading, unloading, transfer and precision assembly of engine components, body sheet metal parts, electronic components, etc., realizing 24-hour automatic operation, improving production line efficiency and reducing labor costs.



适配型号 Compatible Models

适配 (FR3WML、FR5、FR5WML、FR10、FR16、FR20、FR30)。
Compatible with (FR3WML、FR5、FR5WML、FR10、FR16、FR20、FR30)。

机床上下料应用案例 Application Case of Machine Tool Loading and Unloading



协作机器人搭载移动小车，集成视觉定位与抓取夹具，打造灵活高效的机床自动上下料解决方案。

核心优势: 无需固定安装，可快速部署至不同工位；协作机器人保障人机协同安全，柔性适配多种规格工件，实现高效自动化上下料。

典型场景: 适用于 CNC 加工中心、数控车床、磨床等机床的自动上下料作业，可替代人工完成工件取放、装夹与转运，大幅提升生产效率与设备稼动率，降低人工成本。

The collaborative robot is mounted on a mobile cart, integrating visual positioning and grippers, to create a flexible and efficient automatic machine tool loading and unloading solution.

Core Advantages: No fixed installation is required, enabling rapid deployment to different workstations; the collaborative robot ensures human-machine collaborative safety, flexibly adapts to workpieces of various specifications, and realizes efficient automatic loading and unloading.

Typical Scenarios: Suitable for automatic loading and unloading operations of CNC machining centers, CNC lathes, grinding machines and other machine tools. It can replace manual work to complete workpiece pick-and-place, clamping and transfer, greatly improving production efficiency and equipment utilization rate, and reducing labor costs.



适配型号 Compatible Models

适配 (FR5、FR5WML、FR10)。
Compatible with (FR5, FR5WML, FR10).

智能喷涂机器人应用案例 Application Case of Intelligent Spraying Robot



倒挂式协作喷涂机器人，集成视觉定位系统与高效雾化喷涂枪，通过人工手持示教手柄示范喷涂路径，实现高精度、柔性化的自动喷涂作业。

核心优势: 支持人工手持示教，可快速完成复杂曲面喷涂路径的示范与复刻；视觉系统辅助工件精准对位，协作模式保障人机协同安全，大幅降低编程门槛，实现高效自动化喷涂。

典型场景: 适用于工程机械、厨卫洁具、家具板材、钣金构件等行业的工件表面喷涂，可替代人工完成大面积均匀喷涂、边角补喷及异形曲面喷涂，显著提升喷涂效率与涂层质量稳定性。

The inverted collaborative spraying robot integrates a visual positioning system and a high-efficiency atomizing spray gun. It realizes high-precision and flexible automatic spraying operations by demonstrating the spraying path through a manual handheld teach pendant.

Core Advantages: Supports manual handheld teaching, which can quickly complete the demonstration and reproduction of spraying paths for complex curved surfaces. The vision system assists in accurate workpiece positioning, and the collaborative mode ensures human-machine collaborative safety, greatly reducing programming thresholds and achieving efficient automatic spraying.

Typical Scenarios: Suitable for workpiece surface spraying in industries such as engineering machinery, sanitary ware, furniture panels, and sheet metal components. It can replace manual work to complete large-area uniform spraying, corner touch-up spraying, and special-shaped curved surface spraying, significantly improving spraying efficiency and coating quality stability.



适配型号 Compatible Models

适配 (FR16、FR20、FR30)。
Compatible with (FR16、FR20、FR30)。

协作机器人打磨场景应用案例 Application Case of Collaborative Robot in Grinding Scenario



协作机器人集成高负载轻量化关节模组、柔性打磨头、智能感知系统等核心模块，可在木工、金属加工、汽车零部件等场景中完成高精度打磨、抛光作业，实现产线自动化升级。

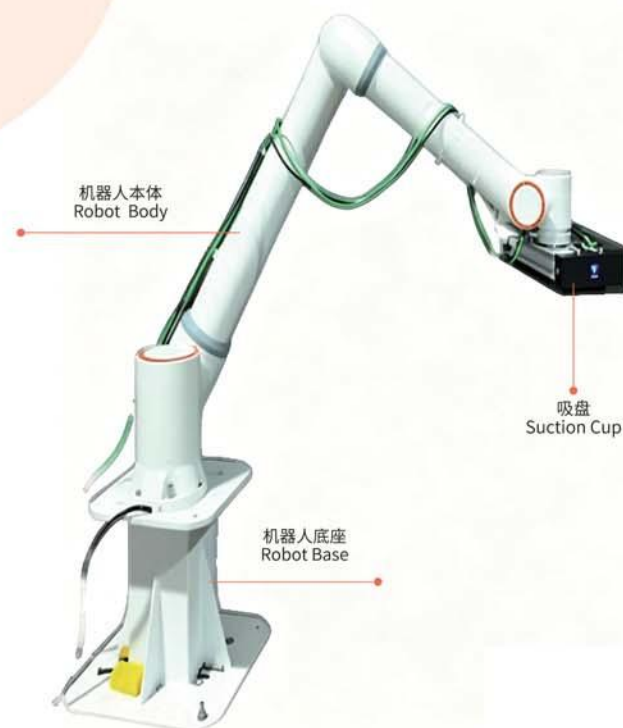
核心优势：搭载高负载轻量化关节模组，兼顾性能与便捷；机械臂灵活高效，运动覆盖广；集成智能感知与控制系统，可精准识别工件、自主规划打磨路径，模块化设计支持快速定制，适配不同行业打磨工艺节拍。

典型场景：适用于木制品打磨、金属件去毛刺、汽车零部件表面抛光等场景，可实现 24 小时自动化作业，提升打磨精度与一致性，降低人工成本与劳动强度。

The collaborative robot integrates core modules such as high-load lightweight joint modules, flexible grinding heads, and intelligent perception systems, enabling high-precision grinding and polishing operations in scenarios like woodworking, metal processing, and automotive parts manufacturing, realizing production line automation upgrading.

Core Advantages: Equipped with high-load lightweight joint modules, balancing performance and convenience; the robotic arm is flexible and efficient with wide motion coverage; integrates intelligent perception and control systems, enabling accurate workpiece recognition and autonomous grinding path planning, with modular design supporting rapid customization to adapt to grinding process rhythms of different industries.

Typical Scenarios: Suitable for scenarios such as wood product grinding, metal part deburring, and automotive parts surface polishing, realizing 24-hour automatic operation, improving grinding accuracy and consistency, and reducing labor costs and intensity.



适配型号 Compatible Models

适配 (FR16、FR20、FR30)。
Compatible with (FR16、FR20、FR30).

3D 视觉码垛搬运应用案例 3D Vision Palletizing & Handling Application Case



采用 3D 视觉拍照扫描识别方案，人工无需繁琐示教编程，设备通过 3D 视觉相机实时扫描工件外形、位置与姿态，自动抓取混料、无序摆放的物料。

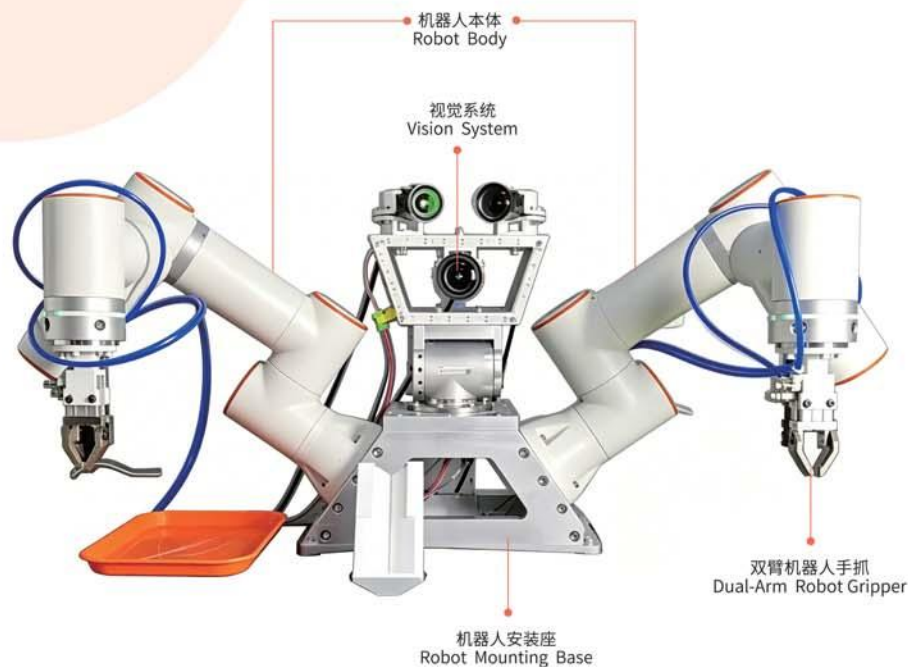
核心优势：3D 视觉全自动扫描建模，无需人工摆放规整；可识别错位、堆叠、杂乱堆放物料，自动规划最优抓取路径；替代人工重复性搬运码垛，降低劳动强度，提升产线自动化与堆放整齐度。

典型场景：广泛应用于仓储物流、食品化工、建材行业、家电制造、汽配工厂等，实现生产线末端成品码垛、来料拆垛、车间物料自动搬运转运。

Adopt 3D vision camera scanning and recognition solution. Complicated manual teaching and programming are not required. The equipment uses a 3D vision camera to scan the workpiece shape, position and attitude in real time, and automatically grab mixed and randomly placed materials.

Core Advantages: Full-automatic 3D vision scanning and modeling, no need for manual regular placement; it can identify misplaced, stacked and disorderly materials, and automatically plan the optimal grabbing path; it replaces manual repetitive handling and palletizing, reduces labor intensity, and improves production line automation and stacking uniformity.

Typical Scenarios: Widely used in warehousing and logistics, food and chemical industry, building materials industry, home appliance manufacturing and auto parts factories, realizing finished product palletizing at the end of production line, depalletizing of incoming materials, and automatic handling and transfer of workshop materials.



适配型号 Compatible Models

适配 FR3、FR5（高精度末端 360°）、FR10、FR16、FR20。
Compatible with FR3、FR5（High-precision End360°）、FR10、FR16、FR20。

双臂机器人 3D 视觉线缆穿孔应用案例 Application Case of Dual-Arm Robot 3D Vision Cable Threading



双臂机器人 3D 视觉线缆穿孔系统，采用高精度 3D 视觉识别与双臂协同控制技术，实现线缆精准穿入微小孔径的自动化作业方案

核心优势：3D 视觉实时定位线缆与孔径位置，毫米级定位精度保障穿孔成功率；双臂协同动作，替代人工精细穿线作业，提升作业效率与稳定性；可适配不同规格线缆与孔径，柔性化满足多样生产需求。

典型场景：适用于电子制造、汽车线束、精密仪器、医疗设备等行业的线缆自动穿孔、穿管作业，缩短装配周期，提升产品一致性与生产自动化水平。

The dual-arm robot 3D vision cable threading system adopts high-precision 3D vision recognition and dual-arm cooperative control technology, realizing an automatic operation solution for accurately threading cables into small holes.

Core Advantages: 3D vision enables real-time positioning of cables and hole positions, millimeter-level positioning accuracy ensures threading success rate; dual-arm cooperative actions replace manual fine threading operations, improving operation efficiency and stability; adaptable to different specifications of cables and hole diameters, flexibly meeting diverse production needs.

Typical Scenarios: Suitable for automatic cable threading and pipe threading operations in electronic manufacturing, automotive wiring harness, precision instruments, medical equipment and other industries, shortening assembly cycle, improving product consistency and production automation level.



适配型号 Compatible Models

适配 FR3WML、FR5、FR5WML。
Compatible with (FR3WML、FR5、FR5WML)。

3D 视觉巡检小车应用案例 Application Case of 3D Vision Inspection Mobile Robot



3D 视觉巡检小车，集成 ALR Lab 智能算法平台与高精度 3D 视觉识别、自主导航技术，可在工业现场、仓储环境等场景中实现自动巡检、缺陷检测与环境感知。

核心优势：结合 ALR Lab 算法，精准识别设备异常、物料偏差与环境隐患；依托 ALR Lab 路径规划能力，替代人工高频巡检，提升效率与数据可靠性；ALR Lab 支持数据可视化与告警，助力产线智能化管理。

典型场景：适用于电力机房、智能制造车间、仓储物流中心等场景的设备巡检、缺陷检测与安全排查，通过 ALR Lab+3D 视觉实现 24 小时自动化巡检，降本增效。

The 3D vision inspection mobile robot integrates the ALR Lab intelligent algorithm platform, high-precision 3D vision recognition and autonomous navigation technology, enabling automatic inspection, defect detection and environment perception in industrial sites and warehouses.

Core Advantages: Combined with ALR Lab algorithms, it accurately identifies equipment anomalies, material deviations and potential hazards; relies on ALR Lab path planning to replace manual high-frequency inspections, improving efficiency and data reliability; ALR Lab supports data visualization and alarms, facilitating intelligent production line management.

Typical Scenarios: Suitable for equipment inspection, defect detection and safety investigation in power rooms, smart workshops and logistics centers. The ALR Lab + 3D vision solution enables 24/7 automatic inspection, reducing costs and improving efficiency.



机器人型号
Robot Model

ART3-R7

人形机器人应用案例 Application Case of Humanoid Robot

人形整机示范 - 全场景作业解决方案

核心组成

1. 半人形动作与交互执行单元
2. 可调节升降柱, 灵活拓展作业高度
3. 移动作业平台, 扩大作业覆盖范围

定制化服务

- 根据场景需求, 定制硬件配置与软件功能
- 各模块接口兼容, 支持快速兼容第三方设备, 降低开发周期与成本



人形机器人集成关节模组、人形机械臂、半人形执行单元、移动作业平台等核心模块, 可在工业制造、巡检运维等场景中实现高精度自动化作业。

核心优势: 搭载高负载轻量化关节模组, 兼顾性能与便捷; 7 自由度人形机械臂灵活高效, 支持 10kg 瞬时负载与 $\pm 0.5\text{mm}$ 精度; 集成多维度感知模块与智能控制系统, 可精准识别环境、自主规划路径, 模块化设计支持快速定制, 助力场景落地。

典型场景: 适用于智能制造车间、电力机房、仓储物流中心等场景, 可完成设备巡检、精密装配、物料搬运等任务, 实现 24 小时自动化作业, 降本增效。

The humanoid robot integrates core modules such as joint modules, humanoid robotic arms, semi-humanoid execution units, and mobile working platforms, enabling high-precision automated operations in industrial manufacturing, inspection and maintenance scenarios.

Core Advantages: Equipped with high-load lightweight joint modules, balancing performance and convenience; 7-DOF humanoid robotic arm is flexible and efficient, supporting 10kg instantaneous load and $\pm 0.5\text{mm}$ accuracy; integrates multi-dimensional perception modules and intelligent control systems, enabling accurate environment recognition and autonomous path planning, with modular design supporting rapid customization to facilitate scenario implementation.

Typical Scenarios: Suitable for smart manufacturing workshops, power equipment rooms, warehouse logistics centers and other scenarios, completing tasks such as equipment inspection, precision assembly, and material handling, realizing 24-hour automatic operation to reduce costs and improve efficiency.